

The objective of this lab was to design and simulate an automated cloud security monitoring and response system using AWS services such as CloudTrail, CloudWatch, EventBridge, Lambda, IAM, and SNS. The system was designed to detect suspicious activities, trigger alerts, and automatically revoke user permissions through a Lambda-based kill-switch mechanism.

The simulation was practically successful. EventBridge and Lambda integration worked successfully, and IAM policy revocation was tested through the Lambda function. CloudTrail logs and EventBridge rules were configured to support automated monitoring and incident response.

Phase 1: INFRASTRUCTURE & LOGGING

Task 1.1 I created a new multi region AWS CloudTrail with a new S3.

cloud Computing Services - Amazon Web Services (A

United States (N. Virginia)

ALIGE OYE (489535778664)

CloudTrail > Trails

Trail successfully created

Trails

Copy events to Lake

Delete

Create trail

	Name	Home region	Multi-region trail	ARN	Insights	Organization trail	S3 bucket	Log file prefix	CloudWatch Logs log group	Status
☐	OyeSecurityMonitor	US East (N. Virginia)	Yes	arn:aws:cloudtrail:us-east-1:489535778664:trail/OyeSecurityMonitor	Disabled	No	oye-cloudtrail-logs-489535778664-6de392c7	-	arn:aws:logs:us-east-1:489535778664:log-group:oye-cloudtrail-logs-489535778664-f9f1db53*	Logging

Below is a history of event showing CloudTrail is being integrated with CloudWatch Log.

AWS CloudTrail Lake feature will no longer be open to new customers starting on May 31, 2026. For capabilities similar to AWS CloudTrail Lake feature explore Amazon CloudWatch. [Learn more](#)

Event history (50+) Info

Event history shows you the last 90 days of management events.

Lookup attributes

Read-only

false

Filter by date and time

Clear filter

< 1 2 ... >

	Event name	Event time	User name	Event source	Resource type	Resource name
☐	CreateLogStream	May 06, 2026, 23:10:22 (UTC+0...	CLOUDWATCH_LO...	logs.amazonaws.com	-	-
☐	CreateLogStream	May 06, 2026, 23:09:42 (UTC+0...	CLOUDWATCH_LO...	logs.amazonaws.com	-	-
☐	StartLogging	May 06, 2026, 23:04:42 (UTC+0...	root	cloudtrail.amazonaws.com	AWS::CloudTrail::Trail	arn:aws:cloudtrail:us-east-1:48...
☐	UpdateTrail	May 06, 2026, 23:04:41 (UTC+0...	root	cloudtrail.amazonaws.com	AWS::CloudTrail::Trail, ...	OyeSecurityMonitor, arn:aws:cl...
☐	CreateLogStream	May 06, 2026, 23:04:41 (UTC+0...	CloudTrail	logs.amazonaws.com	-	-
☐	UpdateTrail	May 06, 2026, 23:04:39 (UTC+0...	root	cloudtrail.amazonaws.com	AWS::CloudTrail::Trail, ...	arn:aws:cloudtrail:us-east-1:48...
☐	UpdateTrail	May 06, 2026, 23:04:35 (UTC+0...	root	cloudtrail.amazonaws.com	AWS::CloudTrail::Trail, ...	arn:aws:cloudtrail:us-east-1:48...

0 / 5 events selected

Task 1.2 Honeytoken

I created a secret in AWS secret manager, this will charge \$0.40 per month

Secret type

Secret type
Other type of secret

Encryption key
aws/secretsmanager

Secret configuration

Secret name
Production_Database_Credentials

Description
Honeytoken secret for oyecloud security monitoring

Tags
-

Resource permissions
-

Secret replication
Disabled

Name of the secret is Production_Database_Credentials

Task1.3 Data residency.

I created a new bucket, and enabled server access logging.

Successfully edited server access logging.

No configurations to display.
[Create configuration](#)

Server access logging [Edit](#)

Log requests for access to your bucket. Use [CloudWatch](#) to check the health of your server access logging. [Learn more](#)

Server access logging
Enabled

Log object key format
[YYYY]-[MM]-[DD]-[hh]-[mm]-[ss]-[UniqueString]

Destination bucket
[s3://exam-storage-bucket](#)

AWS CloudTrail data events

[You can view and configure CloudTrail data events for Amazon S3 bucket object-level operations in the AWS CloudTrail console.](#) [AWS CloudTrail](#)

Phase 2:DETECTION LOGIC FLOW 1

Task 2.1 Metric Filters

I navigated to CloudWatch-logs-group logs and created a new metric filter

A green notification bar at the top states: "Metric filter 'GetSecretValueDetectionFilter' has been created."

The configuration details for the metric filter are as follows:

- Filter pattern:** `{ ($.eventName = "GetSecretValue") && ($.requestParameters.secretId = "Production_Database_Credentials") }`
- Field selection criteria:** -
- Metric:** [SecurityMonitoring](#) / [SecretAccessDetected](#)
- Metric value:** 1
- Default value:** -
- Applied on transformed logs:** -
- Unit:** -
- Emit system field dimensions:** -
- Dimensions:** -

Task 2.2 The threshold

I created a new alarm and did the configuration, set the alarm to trigger in one minute if secret is accessed.

A green notification bar at the top states: "Successfully created alarm eye-exam-alarm." with a "View alarm" button.

The "Alarms" tab is selected, showing "Mute rules - new".

Alarms (1)

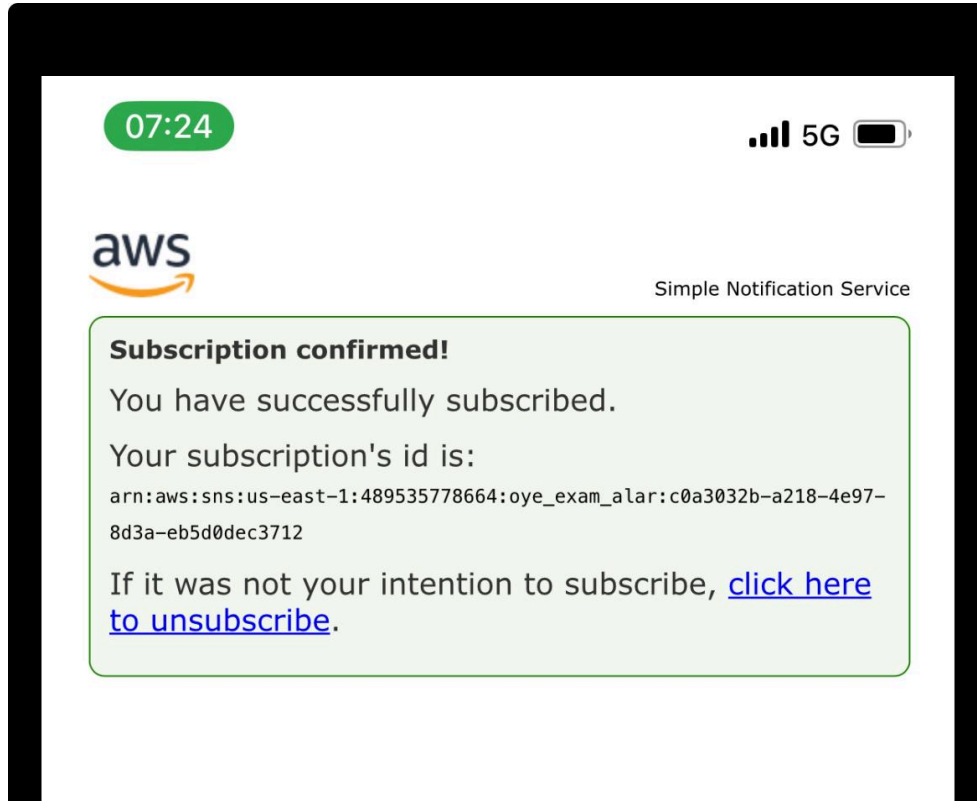
Buttons: Clear selection, Create composite alarm, Actions, Create alarm

Search: Filter alarms, Quick filters

	Name	State	Actions	Actions muted - new	Last state update (UTC)	Conditions
☐	eye-exam-alarm	Insufficient data	Actions enabled	-	2026-05-07 06:27:53	SecretAccessDetect within 1 minute

Task 2.3 Notification

I link the alarm to my SNS Topic and subscribe my student email to receive notifications



PHASE 3 DETECTION LOGIC FLOW 2

Task 3.1 Event-Driven Security

I created a rule name SensitiveDataAccessRule using an Amazon EventBridge that triggers whenever a sensitive data access event occurs in CloudTrail.

Find rules

Any status

<

1

>

	Name	Status	Type	Event bus	ARN	Description
	SensitiveDataAccessRule	Enabled	Standard	default	arn:aws:events:us-east-1:489535778664:rule/SensitiveDataAccessRule	-

SensitiveDataAccessRule

[Edit](#)
[Disable](#)
[Delete](#)
[CloudFormation Template ▾](#)

Rule details [Info](#)

Rule name
SensitiveDataAccessRule

Status
Enabled

Event bus name
default

Type
Standard

Description

-

Rule ARN

arn:aws:events:us-east-1:489535778664:rule/SensitiveDataAccessRule

Event bus ARN

arn:aws:events:us-east-1:489535778664:event-bus/default

Event pattern

Targets

Monitoring

Tags

Event pattern [Info](#)

[Edit](#)

```
1 {
2   "source": ["aws.secretsmanager"],
3   "detail-type": ["AWS API Call via CloudTrail"],
4   "detail": {
5     "eventSource": ["secretsmanager.amazonaws.com"],
6     "eventName": ["GetSecretValue"]
7   }
8 }
```

[Copy](#)

Task 3.2 Comparison

SNS notification(seperate topic) I created a new topic

Topic SensitiveDataAccessTopic2 created successfully.

You can create subscriptions and send messages to them from this topic.

[Publish message](#)

✕

Notifications

0

0

1

1

0

0

▾

SensitiveDataAccessTopic2

[Edit](#)
[Delete](#)
[Publish message](#)

Details

Name
SensitiveDataAccessTopic2

ARN
arn:aws:sns:us-east-1:489535778664:SensitiveDataAccessTopic2

Display name
SecurityAlert2

Type
Standard

Topic owner
489535778664

Subscriptions

Access policy

Delivery policy (HTTP/S)

Delivery status logging

Encryption

Tags

Integrations

Subscriptions (0)

[Edit](#)
[Delete](#)
[Request confirmation](#)
[Confirm subscription](#)
[Create subscription](#)

<

1

>

⚙

ID	Endpoint	Status	Protocol
----	----------	--------	----------

No subscriptions found

You don't have any subscriptions to this topic.

From the topic I created and added a subscription

✔ Subscription to SensitiveDataAccessTopic2 created successfully.
The ARN of the subscription is arn:aws:sns:us-east-1:489535778664:SensitiveDataAccessTopic2:6d907134-837b-4094-a2c2-193c3b95264d.

Notifications 0 0 1 1 0

Subscription: 6d907134-837b-4094-a2c2-193c3b95264d

Details

ARN
arn:aws:sns:us-east-1:489535778664:SensitiveDataAccessTopic2:6d907134-837b-4094-a2c2-193c3b95264d

Endpoint
oyealige@gmail.com

Topic
[SensitiveDataAccessTopic2](#)

Status
⌚ Pending confirmation

Protocol
EMAIL

Subscription filter policy

Redrive policy (dead-letter queue)

Subscription filter policy [Info](#)
This policy filters the messages that a subscriber receives.

No filter policy configured for this subscription.
To apply a filter policy, edit this subscription.

09:25



Simple Notification Service

Subscription confirmed!

You have successfully subscribed.

Your subscription's id is:

arn:aws:sns:us-east-1:489535778664:SensitiveDataAccessTopic2:6d907134-837b-4094-a2c2-193c3b95264d

If it was not your intention to subscribe, [click here to unsubscribe](#).

Task 3.3 THE BRIDGE

At the AWS CLI this code was run “aws secretsmanager get-secret-value --secret-id Production_Database_Credentials”

Below is the output

A screenshot of the AWS CloudShell interface. The top bar shows the AWS logo, a search bar, and a keyboard shortcut [Alt+S]. Below the bar, the 'CloudShell' title is visible. The terminal window shows the command 'aws secretsmanager get-secret-value --secret-id Production_Database_Credentials' being executed. The output is a JSON object containing details about the secret, including its ARN, name, version ID, secret string (formatted with escape characters), version stages, and creation date.

```
~ $ aws secretsmanager get-secret-value --secret-id Production_Database_Credentials
{
  "ARN": "arn:aws:secretsmanager:us-east-1:489535778664:secret:Production_Database_Credentials-wf8wHg",
  "Name": "Production_Database_Credentials",
  "VersionId": "bb2b523a-fd29-423d-b821-59fa6f4d53fd",
  "SecretString": "{\r\n  \"username\": \"admin\", \r\n  \"password\": \"SuperSecret123!\", \r\n}",
  "VersionStages": [
    "AWSCURRENT"
  ],
  "CreateDate": "2026-05-06T22:53:09.002000+00:00"
}
```

```
{
  "ARN":
"arn:aws:secretsmanager:us-east-1:489535778664:secret:Production_Database_Credentials-wf8wHg",
  "Name": "Production_Database_Credentials",
  "VersionId": "bb2b523a-fd29-423d-b821-59fa6f4d53fd",
  "SecretString": "{\r\n  \"username\": \"admin\", \r\n  \"password\": \"SuperSecret123!\", \r\n}",
  "VersionStages": [
    "AWSCURRENT"
  ],
  "CreateDate": "2026-05-06T22:53:09.002000+00:00"
}
```

PHASE 4: THE “ACTIVE DEFENSE” KILL-SWITCH

TASK 4.1

I created a simple AWS lambda function using python 3.11, it uses AdminDetachUserPolicy to strip all permissions from a user. IAM permission is being given and deployed in JSON

Successfully created the function "DisableUserPermissionsFunction". You can now change its code and configuration. To invoke your function with a test event, choose "Test".

DisableUserPermissionsFunction

Throttle Copy ARN Actions

Export to Infrastructure Composer Download

Function overview Info

Diagram Template

DisableUserPermissionsFunction

Layers (0)

+ Add trigger + Add destination

Function ARN
arn:aws:lambda:us-east-1:489535778664:function:DisableUserPermissionsFunction

Description
-

Last modified
35 seconds ago

Code Test Monitor Configuration Aliases Versions

Code source Info Open in Visual Studio Code i Update

TASK 4.2 THE TRIGGER

I have connected a trigger(EventBridge Rule) to the lambda function

DisableUserPermissionsFunction

✓ The trigger SensitiveDataAccessRule was successfully added to function DisableUserPermissionsFunction.

Function overview Info

Diagram Template

DisableUserPermissionsFunction

Layers (0)

EventBridge (CloudWatch Events)

+ Add trigger + Add destination

Code Test Monitor Configuration Aliases Versions

TASK 4.3 THE "TRAP" TEST

- 1) Victim IAM user created, with [AWS SecretsManagerClientReadOnlyAccess](#) (NOTE: There is no "SecretsManagerRead" instead there is

“[AWS Secrets Manager ClientReadOnlyAccess](#)”)

> victimuser

1

5
)

it
requests

victimuser info

Delete

Summary

ARN
arn:aws:iam::489535778664:user/victimuser

Created
May 09, 2026, 09:33 (UTC+01:00)

Console access
Disabled

Last console sign-in
-

Access key 1
[Create access key](#)

Permissions

Groups

Tags

Security credentials

Last Accessed

Permissions policies (1)

Remove

Add permissions

Permissions are defined by policies attached to the user directly or through groups.

Search

Filter by Type
All types

< 1 > ⚙

☐

Policy name

▲

Type

▼

Attached via

☐

[AWSecretsManagerClientReadOnlyAccess](#)

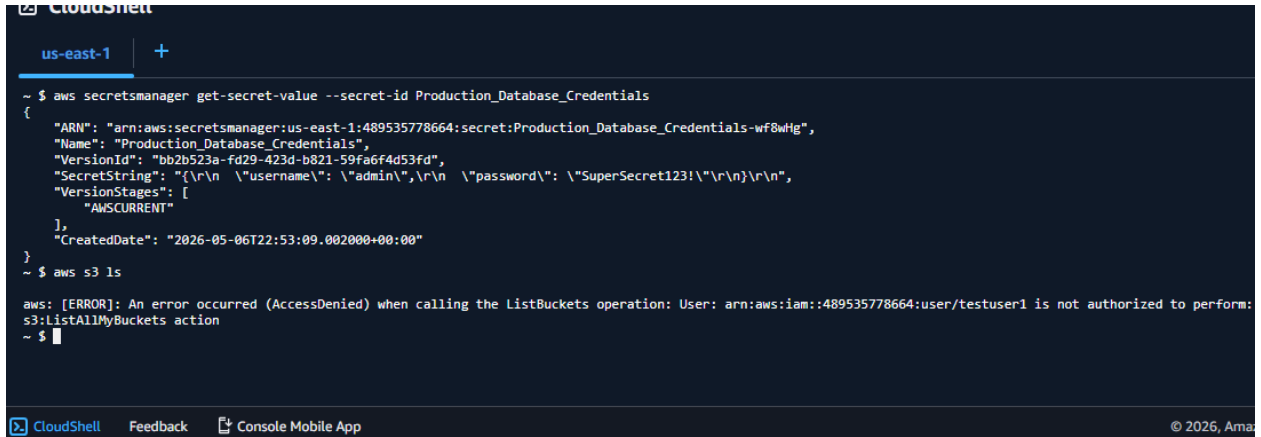
AWS managed

Directly

2) Victim user was used to access the CLI but access was denied

<	3	"userIdentity": {
	4	"type": "IAMUser",
	5	"principalId": "AIDAXD6U3F5UCVH3SEFBB",
	6	"arn": "arn:aws:iam::489535778664:user/testuser1",
	7	"accountId": "489535778664",
	8	"accessKeyId": "AKIAXD6U3F5U06WJKC6W",
	9	"userName": "testuser1"
	10	},
	11	"eventTime": "2026-05-09T23:31:51Z",
	12	"eventSource": "s3.amazonaws.com",
	13	"eventName": "ListBuckets",
	14	"awsRegion": "us-east-1",
	15	"sourceIPAddress": "44.223.40.226",
	16	"userAgent": "[aws-cli/2.34.43 md/awscrt#0.32.2 ua/2.1 os/linux#6.1.166-197.305.amz
:ores	17	"errorCode": "AccessDenied",
	18	"errorMessage": "User: arn:aws:iam::489535778664:user/testuser1 is not authorized t
	19	"requestParameters": {
	20	"Host": "s3.us-east-1.amazonaws.com"
	21	},
	22	"responseElements": null,
	23	"additionalEventData": {
	24	"SignatureVersion": "SigV4",
	25	"CipherSuite": "TLS_AES_128_GCM_SHA256",
	26	"bytesTransferredIn": 0,
↗	27	"AuthenticationMethod": "AuthHeader",
	28	"x-amz-id-2": "RH5c1f/cGaUvucbzCX7dLgzQsEe4Z5RoAGSd0onTFI8F3CJGZ4/x0uzfnkfZqC39
	29	"bytesTransferredOut": 0

3) The Result: Access was revoked after second attempt to run another command on CLI
Snapshot of access been revoked



```
CloudShell
us-east-1 +
~ $ aws secretsmanager get-secret-value --secret-id Production_Database_Credentials
{
  "ARN": "arn:aws:secretsmanager:us-east-1:489535778664:secret:Production_Database_Credentials-wf8wHg",
  "Name": "Production_Database_Credentials",
  "VersionId": "bb2b523a-fd29-423d-b021-59fa6f4d53fd",
  "SecretString": "{\r\n  \"username\": \"admin\", \r\n  \"password\": \"SuperSecret123!\", \r\n}\r\n",
  "VersionStages": [
    "AMSCURRENT"
  ],
  "CreateDate": "2026-05-06T22:53:09.002000+00:00"
}
~ $ aws s3 ls
aws: [ERROR]: An error occurred (AccessDenied) when calling the ListBuckets operation: User: arn:aws:iam::489535778664:user:testuser1 is not authorized to perform: s3:ListAllMyBuckets action
~ $
```

CloudShell Feedback Console Mobile App © 2026, Amazon

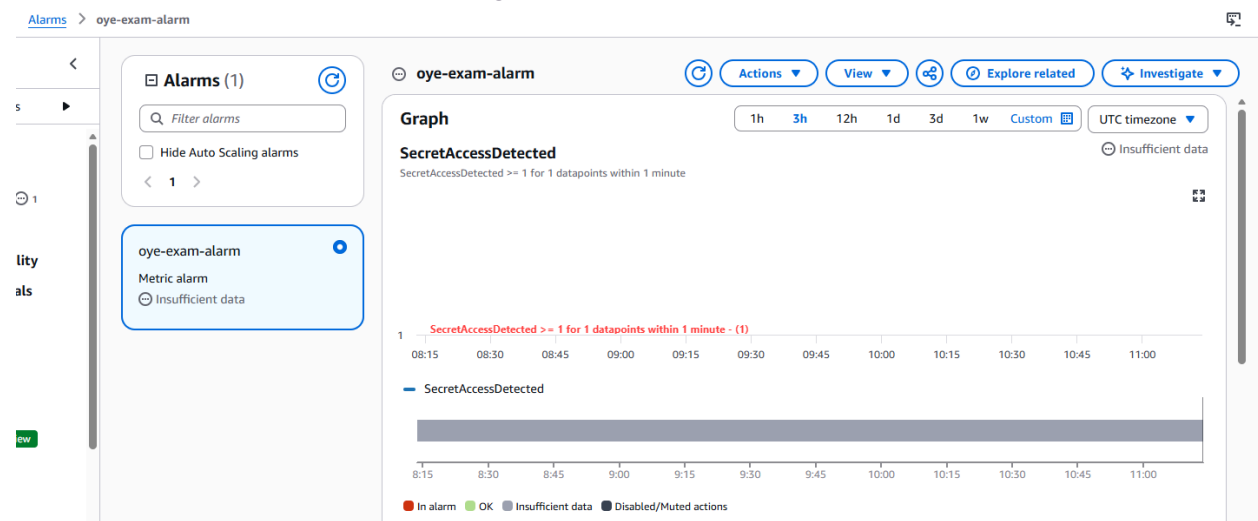
PHASE 5: FORENSIC REPORT

1) Executive Summary

The objective of this lab was to design and simulate an automated cloud security monitoring and response system using AWS services such as CloudTrail, CloudWatch, EventBridge, Lambda, IAM, and SNS. The system was designed to detect suspicious activities, trigger alerts, and automatically revoke user permissions through a Lambda-based kill-switch mechanism. The simulation was successful. EventBridge and Lambda integration worked successfully, and IAM policy revocation was tested through the Lambda function. CloudTrail logs and EventBridge rules were configured to support automated monitoring and incident response.

2) Lab Architecture

A Picture of CloudWatch Alarm Configuration



Screenshot of EventBridge Rule below;

Amazon EventBridge > Rules > SensitiveDataAccessRule

Dashboard

Developer resources

Learn

Sandbox

Quick starts

Buses

Event buses

Rules Updated

Global endpoints

Archives

Replays

Pipes

Pipes

Scheduler

Schedules

Schedule groups

Scheduled rules (legacy) Updated

Integration

Partner event sources

API destinations

SensitiveDataAccessRule

EditDisableDeleteClose

Rule details Info

Rule name

SensitiveDataAccessRule

Description

-

Status

Enabled

Rule ARN

arn:aws:events:us-east-1:4895357786:rule/SensitiveDataAccessRule

Event bus name

default

Event bus ARN

arn:aws:events:us-east-1:4895357786:event-bus/default

Type

Standard

Event pattern

Targets

Monitoring

Tags

Event pattern Info

```
1 {
2   "source": ["aws.secretsmanager"],
3   "detail-type": ["AWS API Call via CloudTrail"],
4   "detail": {
5     "eventSource": ["secretsmanager.amazonaws.com"],
6     "eventName": ["GetSecretValue"]
7   }
8 }
```

Copy

3) The Evidence Log

Event Time:May 09, 2026, 23:41:08 (UTC+01:00)

User Identity:testuser1

Source IP address:3.238.163.94

CloudTrail > Event history > DetachUserPolicy

CloudTrail

<

Dashboard

Event coverage

Event history

Query

Event data stores

Integrations

Settings

Help

Documentation

APIs

SDKs

18

2026-05-09T22:29:43Z",

19

"mfaAuthenticated": "false"

20

}

21

},

22

"inScopeOf": {

23

"issuerType": "AWS::Lambda::Function",

24

"credentialsIssuedTo": "arn:aws:lambda:us-east-1:489535778664:function:DisableUs

25

}

26

},

27

"eventTime": "2026-05-09T22:41:08Z",

28

"eventSource": "iam.amazonaws.com",

29

"eventName": "DetachUserPolicy",

30

"awsRegion": "us-east-1",

31

"sourceIPAddress": "3.238.163.94",

32

"userAgent": "Boto3/1.40.4 md/Botocore#1.40.4 ua/2.1 os/linux#5.10.252-285.992.amzn2.x86

33

"requestParameters": {

34

"userName": "testuser1",

35

"policyArn": "arn:aws:iam::aws:policy/AWSSecretsManagerClientReadOnlyAccess"

36

},

37

"responseElements": null,

38

"requestID": "95e29fdd-d917-439e-9d5b-ec653270d467",

39

"eventID": "b46b8f97-f1f4-40e3-b09f-ea492634ae5f",

40

"readOnly": false,

41

"eventType": "AwsApiCall",

42

"managementEvent": true,

43

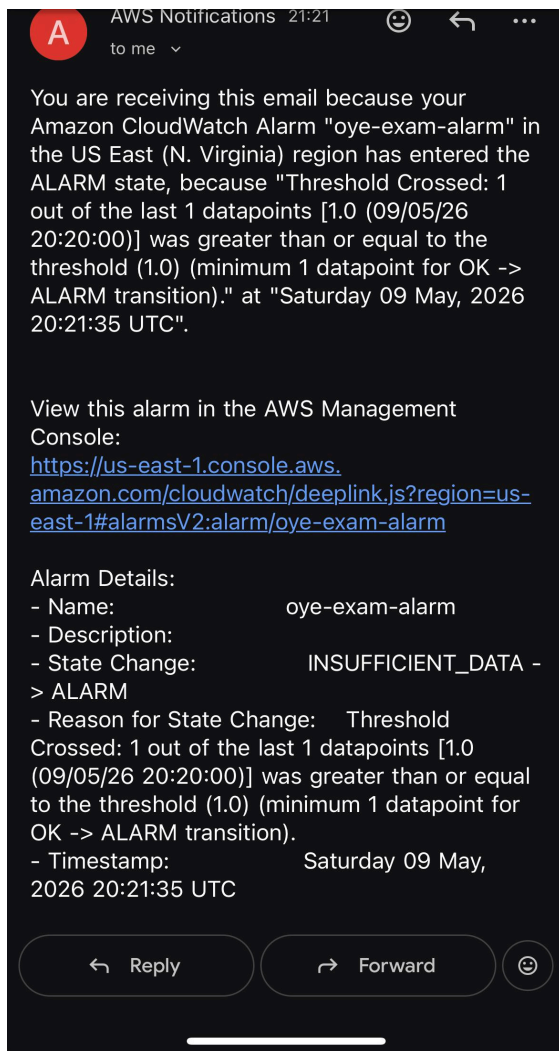
"recipientAccountId": "489535778664",

44

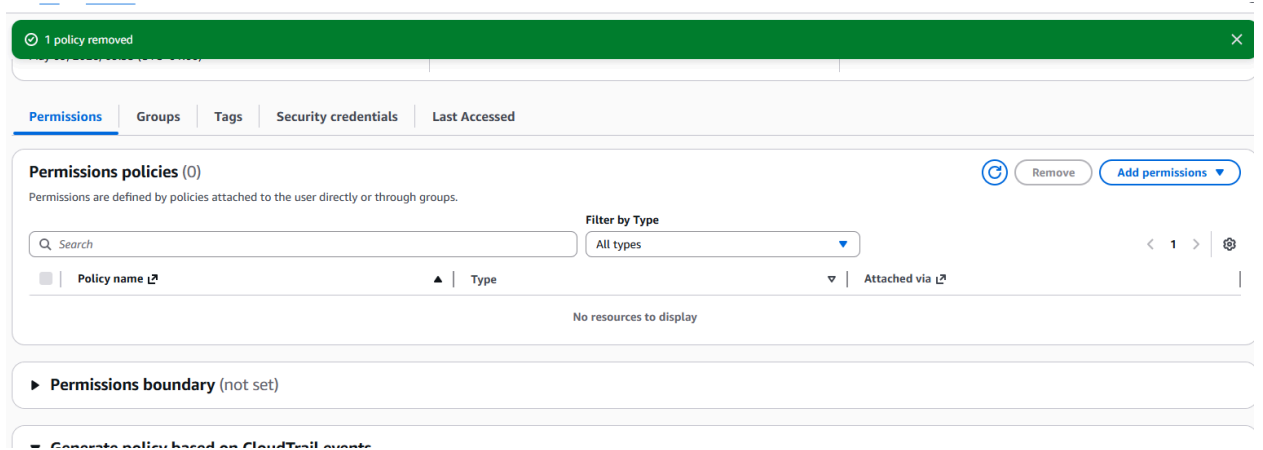
"eventCategory": "Management"

4) Technical Evidence:

- Screenshot of email notification from flow 1



- The Kill-Switch mechanism was validated by removing the attached IAM permissions from the victim user account. After revocation, the user no longer retained the SecretsManagerReadOnly policy. •
OBSERVATION: After the revocation I tried adding the readonly permission to the IAM user again but it wasn't successful.



- The Log:Snippet of CloudTrail JSON Error code showing AccessDenied for the users second attempt using the CLI.

```

3      "userIdentity": {
4          "type": "IAMUser",
5          "principalId": "AIDAXD6U3F5UCVH3SEFBB",
6          "arn": "arn:aws:iam::489535778664:user/testuser1",
7          "accountId": "489535778664",
8          "accessKeyId": "AKIAXD6U3F5U06WJKC6W",
9          "userName": "testuser1"
10     },
11     "eventTime": "2026-05-09T23:31:51Z",
12     "eventSource": "s3.amazonaws.com",
13     "eventName": "ListBuckets",
14     "awsRegion": "us-east-1",
15     "sourceIPAddress": "44.223.40.226",
16     "userAgent": "[aws-cli/2.34.43 md/awscrt#0.32.2 ua/2.1 os/linux#6.1.166-197.305.amz",
17     "errorCode": "AccessDenied",
18     "errorMessage": "User: arn:aws:iam::489535778664:user/testuser1 is not authorized to",
19     "requestParameters": {
20         "Host": "s3.us-east-1.amazonaws.com"
21     },
22     "responseElements": null,
23     "additionalEventData": {
24         "SignatureVersion": "SigV4",
25         "CipherSuite": "TLS_AES_128_GCM_SHA256",
26         "bytesTransferredIn": 0,
27         "AuthenticationMethod": "AuthHeader",
28         "x-amz-id-2": "RH5clf/cGaUvucbzCX7dLgzQsEe4Z5RoAGSd0onTFI8F3CJGZ4/x0uzfnkfZqC39",
29         "x-amz-token": "..."

```

5) Security Analysis (The Spicy Part)

- i) Speed: Flow 2 (EventBridge) provided faster notification and response compared to Flow 1 (CloudWatch Metric Filter). EventBridge reacts directly to AWS events in near real time, while CloudWatch metric filters depend on log ingestion and metric evaluation intervals.
- ii) Context: The EventBridge alert provided more contextual information because it included details such as the IAM username, API action, and source IP address. This made investigation easier and improved incident response efficiency.
- iii) Compliance: This lab supports the Continuous Monitoring requirement of frameworks such as NIST because it continuously monitors cloud activity using CloudTrail, analyzes security events using EventBridge and CloudWatch, and automatically responds to suspicious behavior through Lambda automation.

6) If this system were to be improved for use in a banking environment, EventBridge (Flow 2) would be preferred over CloudWatch Metric Filters (Flow 1) for triggering the Kill-Switch mechanism. During the lab, EventBridge provided faster and more flexible event-driven automation. It was able to react directly to CloudTrail events and trigger the Lambda function in near real time.

EventBridge also provided richer event details such as the API action, IAM username, and source IP address, which improved investigation and response accuracy. In contrast, CloudWatch Metric Filters depended on log processing and metric evaluation, making the response slower and less detailed.

Based on the lab experience, EventBridge is more suitable for banking environments where rapid detection and immediate automated response are critical for protecting sensitive financial systems and customer data.